

Application Layer – Question List by Marks

The **Application Layer** is the top layer of the TCP/IP model and OSI model. Exam questions usually focus on **DNS, HTTP, FTP, SMTP, POP3, IMAP, Telnet, SSH, SNMP, DHCP, and Application Layer services.**

5 Marks Questions

Basics of Application Layer

1. Define the Application Layer.
2. List the functions of the Application Layer.
3. What are Application Layer protocols?
4. Differentiate between Application Layer and Transport Layer.
5. What are network applications?

DNS

6. What is DNS?
7. What is a domain name?
8. What is DNS resolution?
9. What is a DNS server?
10. What are the advantages of DNS?

HTTP and Web Services

11. What is HTTP?
12. What is HTTPS?
13. What is a URL?
14. What is a web browser?
15. What is a web server?

FTP

16. What is FTP?
17. What are the features of FTP?
18. What are the advantages of FTP?

Email Protocols

19. What is SMTP?
20. What is POP3?
21. What is IMAP?
22. Differentiate between POP3 and IMAP.

Remote Access

23. What is Telnet?
24. What is SSH?
25. Differentiate between Telnet and SSH.

Network Management

26. What is SNMP?
27. What is network management?

Miscellaneous

28. What is DHCP?
29. What is client-server architecture?
30. What is peer-to-peer architecture?
31. What is a socket?
32. What is a proxy server?
33. What is cloud computing?
34. What is an application service?

10 Marks Questions

Application Layer Concepts

1. Explain the functions of the Application Layer.
2. Explain client-server architecture.
3. Explain peer-to-peer architecture.
4. Compare client-server and peer-to-peer models.
5. Explain network applications and services.

DNS

6. Explain DNS and its working.
7. Explain DNS hierarchy.
8. Explain domain name resolution process.
9. Explain the structure of a domain name.

HTTP and WWW

10. Explain HTTP protocol.
11. Explain HTTP request and response messages.
12. Explain HTTPS and its advantages.
13. Explain the architecture of the World Wide Web.
14. Explain the working of a web browser and web server.

FTP

15. Explain FTP and its working.
16. Explain FTP architecture.
17. Discuss advantages and disadvantages of FTP.

Email Services

18. Explain SMTP protocol.
19. Explain POP3 protocol.
20. Explain IMAP protocol.
21. Compare POP3 and IMAP.
22. Explain the process of sending and receiving email.

Remote Access

23. Explain Telnet protocol.
24. Explain SSH protocol.
25. Compare Telnet and SSH.

Network Management

26. Explain SNMP protocol.
 27. Explain network management using SNMP.
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15 Marks Questions

1. Explain the Application Layer in detail with its functions and services.
2. Explain DNS architecture and the domain name resolution process with a neat diagram.
3. Explain HTTP protocol and HTTP message format.
4. Explain the working of the World Wide Web (WWW).
5. Explain HTTPS and compare it with HTTP.
6. Explain FTP architecture and working with a neat diagram.
7. Explain electronic mail architecture and email protocols.
8. Explain SMTP, POP3, and IMAP in detail.
9. Compare POP3 and IMAP.
10. Explain Telnet and SSH protocols with comparison.
11. Explain client-server architecture and peer-to-peer architecture.
12. Explain SNMP architecture and network management.
13. Explain the process of sending and receiving email through SMTP and POP3/IMAP.
14. Explain Application Layer protocols and their functions.
15. Discuss the role of DNS, HTTP, FTP, and SMTP in Internet communication.
16. Explain web services and Internet applications.
17. Explain remote login and file transfer services.
18. Explain the architecture and operation of DNS servers.
19. Explain Application Layer services provided in the TCP/IP model.
20. Compare major Application Layer protocols.

Frequently Asked Long Questions (15 Marks)

1. Application Layer and its functions.
2. DNS and domain name resolution.
3. HTTP and HTTPS.
4. FTP architecture and working.

5. SMTP, POP3, and IMAP.
 6. Telnet and SSH.
 7. Client-Server and Peer-to-Peer Architecture.
 8. SNMP protocol.
 9. Email architecture.
 10. Application Layer protocols and services.
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Frequently Asked Medium Questions (10 Marks)

1. Explain DNS.
 2. Explain HTTP.
 3. Explain HTTPS.
 4. Explain FTP.
 5. Explain SMTP.
 6. Explain POP3.
 7. Explain IMAP.
 8. Explain SSH.
 9. Explain SNMP.
 10. Compare Client-Server and Peer-to-Peer models.
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Frequently Asked Short Questions (5 Marks)

1. Define Application Layer.
2. Functions of Application Layer.
3. DNS.
4. HTTP.
5. HTTPS.
6. FTP.
7. SMTP.
8. POP3.

9. IMAP.
 10. Telnet.
 11. SSH.
 12. SNMP.
 13. URL.
 14. Web Server.
 15. Web Browser.
 16. Domain Name.
 17. Client-Server Architecture.
 18. Peer-to-Peer Architecture.
 19. Proxy Server.
 20. Cloud Computing.
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Additional Topics Often Included in Modern Syllabi

If your syllabus includes newer Application Layer concepts, add these questions:

5 Marks

- What is a web cookie?
- What is a search engine?
- What is a Content Delivery Network (CDN)?
- What is REST API?
- What is a web service?

10 Marks

- Explain RESTful web services.
- Explain cloud-based application services.
- Explain CDN architecture and benefits.

15 Marks

- Explain modern web applications and Application Layer protocols.
- Explain web services, APIs, and cloud applications.

Most Important Exam Questions

15 Marks

1. DNS and Domain Name Resolution.
2. HTTP and HTTPS.
3. FTP Architecture and Working.
4. SMTP, POP3, and IMAP.
5. Application Layer Functions.
6. Telnet vs SSH.
7. Client-Server vs Peer-to-Peer Architecture.

10 Marks

1. DNS.
2. HTTP.
3. FTP.
4. SMTP.
5. POP3 and IMAP.
6. SNMP.

5 Marks

1. DNS.
2. HTTP.
3. HTTPS.
4. FTP.
5. SMTP.
6. POP3.
7. IMAP.
8. SSH.
9. URL.
10. Domain Name.

